



YOUR FIRST GODOT GAME · AGES 12 & UP

Make a character walk and jump — your first win in Godot

Godot can feel like a cockpit with a hundred buttons. It isn't, really — it's just **four areas** that never move. Learn those four, and the rest is easy. This guide gets you from "I opened Godot" to "my character jumps!" — one small click at a time.

✓ Plain words, no jargon

✓ One action per step

✓ Check-off boxes

✓ Print it as a cheat sheet

✓ Godot 4

The map: 4 areas that never move

Every colour below stays the same for the *whole* guide. When a step says **green**, it means the green area. You never have to remember names — just follow the colours.

▲ Top bar – the menus & the GO button

▶ GO / Play

■ Scene – the PARTS list

Every piece of your game (player, ground, coin) shown as a little tree. Top-left.

■ Files – your STUFF box

All your pictures and sounds live here. Bottom-left.

■ Stage – where you SEE your game

The big middle window. You drag things here and watch it come to life.

Press ▶ (top-right) to play it for real.

■ Inspector – the SETTINGS knobs

Click any part, and its settings (size, speed, picture) appear here. Right side.

Bottom strip = messages from Godot (errors show here in red). Peek only when something breaks.

■ Parts list (Scene)

■ Stage (Viewport)

■ Settings (Inspector)

■ Files (FileSystem)

■ GO (Play)

4 words you'll keep hearing

That's it. Learn these four and you can follow any Godot tutorial.

Node

One single Lego brick. A picture is a node. A sound is a node. A shape is a node.

Think: **a piece.**

Scene

A bunch of nodes clicked together into one thing — like your whole Player.

Think: **a finished build** made of pieces.

Inspector

The settings panel. Click a node, change its numbers and pictures here.

Think: **the knobs.**

Script

A page of instructions that tells a node how to act — like "jump when I press space."

Think: **the brain**.

PART 1

Your first win — walk & jump

Do these in order. Tick each box when you finish it. By the end, a character you made will run and jump when you press the keys. That's a real game moment. 🎉

1

TOP BAR

Open Godot — from scratch, or the camp's project

Open the **Godot** app first. A list of projects appears. Pick your path:

1. **Starting from scratch:** click **+ New Project** (top-right), name it **Reef Runner**, pick a folder, click **Create & Edit**.
2. **The camp gave you a project folder:** click **Import**, find that folder, open the **project.godot** file inside it, then **Import & Edit**.
3. Either way, the big editor opens. Find your four colours from the map. You're home.

You'll know it worked: you see the editor with its 4 areas — an empty dark **Stage** in the middle (from scratch), or the camp's game already loaded.

Make the character (a new Scene)

1. In the **Parts list** (top-left), click **+ Other Node** (or the big "Other Node" button).
2. In the search box, type **CharacterBody2D**. Click it, then **Create**.
3. It appears in the Parts list. **Double-click its name** and rename it **Player**.

Why this one? CharacterBody2D is Godot's special node for "a thing the player controls that bumps into walls and floors." It's the body your hero lives in.

You'll know it worked: **Player** is now at the top of the Parts list.

Give her a picture and a body shape

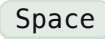
A CharacterBody2D is invisible on its own. It needs two child pieces: something to see, and something to *bump* with.

1. **Right-click Player** → **Add Child Node** → search **Sprite2D** → Create.
(This is her picture.)
2. Click the new **Sprite2D**. In the **Inspector** (right), find **Texture** → **Load** → pick any character image from your **Files**. (No image yet? Godot's built-in *icon.svg* works fine for now.)
3. **Right-click Player** again → **Add Child Node** → search **CollisionShape2D** → Create. (This is her bump-shape.)
4. Click **CollisionShape2D**. In the **Inspector**, click **Shape** → **New RectangleShape2D**. Drag its handles in the **Stage** until the box roughly covers her.

You'll know it worked: Player now has two pieces under it in the Parts list, and you see her picture in the Stage.

Teach Godot the keys (Input Map)

The brain we add next will say things like "when *move_right* is pressed...". First we tell Godot what those names mean. Do this once.

1. Top bar: **Project** → **Project Settings** → tab **Input Map**.
2. In "Add New Action", type **move_left** → **Add**. Do the same for **move_right** and **jump**.
3. Next to **move_left**, click the **+**, press the  Left Arrow key, confirm. For **move_right** use . For **jump** press .
4. Click **Close**.

Spelling matters here. The names must be exactly `move_left`, `move_right`, `jump` — all lowercase, with the underscore. The code won't find them otherwise.

You'll know it worked: three actions are listed with a key icon next to each.

Give her a brain (the movement script)

1. **Right-click** **Player** → **Attach Script** → **Create**. A code page opens.
2. Delete what's there and type (or paste) the code below.
3. Press **Ctrl** / **⌘** + **S** to save.

Player.gd – the movement brain

GDScript

```
extends CharacterBody2D

@export var speed: float = 250.0
@export var jump_force: float = -450.0
@export var gravity: float = 1200.0

@onready var visual = get_node_or_null("Sprite2D")

func _physics_process(delta):
    # Gravity pulls her back down.
    if not is_on_floor():
        velocity.y += gravity * delta

    # Move left and right.
    var direction = Input.get_axis("move_left", "move_right")
    if direction != 0:
        velocity.x = direction * speed
    else:
        velocity.x = move_toward(velocity.x, 0, speed)

    # Jump – only when standing on the ground.
    if Input.is_action_just_pressed("jump") and is_on_floor():
        velocity.y = jump_force

    # Face the way she's walking.
    if direction != 0 and visual:
        visual.flip_h = direction < 0

    move_and_slide()
```

In plain words: every tick, **fall** if not on the floor · **slide** left/right with the arrow keys · **jump** if Space is pressed and she's on the ground · **flip** the picture to face her direction · then **do the move** (move_and_slide).

You'll know it worked: a little scroll icon appears next to Player in the Parts list, and there are no red errors at the bottom.

6

PARTS LIST

Give her ground to stand on

Right now she'd fall forever. Let's add a floor. We'll build it as its own little scene root so the Player has a world to live in.

1. Click the **+** tab at the top of the **Stage** to make a **new scene**. Add a **Node2D**, rename it **Level**.
2. **Right-click** Level → **Add Child Node** → **StaticBody2D**. Add a **CollisionShape2D** under it with a wide, flat **RectangleShape2D** — that's the floor.
3. Add a **Sprite2D** (or ColorRect) so you can see the floor too.
4. Drag your **Player.tscn** from the **Files** panel into the **Level** parts list. Move her up above the floor in the Stage.

You'll know it worked: in the Level scene you see the floor and the Player sitting above it.

7

TOP BAR • GO

Press GO and play! 🎉

1. Press the big **▶ Play** button (top-right) or **F5**.
2. Godot asks for a **main scene** the first time — pick **Level**.
3. Press the **←** **→** arrows to walk. Press **Space** to jump.

🎉 **YOU DID IT.** You made a character move in a game you built. Everything else — coins, enemies, score — is just more of this same loop: add a piece, give it a brain, press GO.

Help! It's not working

This is the normal part of making games — not a sign you did it wrong.
Find your problem, open it, fix it.

😵 She falls straight through the floor ▶

📌 She won't move at all ▶

🚀 She flies up and never comes down ▶

■ Red text at the bottom of the screen ▶

■ Nothing happens when I press Play ▶

PART 2 The full game — build it piece by piece

Once the first win clicks, the rest of the camp game (the "save the reefs" platformer) is the same moves repeated. Each box below matches the **video tutorials in the handout**, in order. Open one at a time — never all at once.

DAY 1 Set up & make the player ▶

DAY 2 Platforms, background & nicer movement ▶

DAY 3 Coins, enemies, score & game-over ▶

DAY 4 • 5 Playtest, present & the comic ▶

Notes for you, the guide

You don't have to remember Godot — you have to remember the **4 colours** and point at them. That's the whole job.

- **Teach the map first, code last.** Spend the first 5 minutes just naming the 4 areas out loud together. Everything after lands easier.
- **One step on screen at a time.** Do a step, let them do it, tick the box, then move on. The check-off boxes keep restless attention anchored.
- **Say the "why" in one line.** "The CollisionShape is her bump-shape."
Kids remember the picture, not the term.
- **Expect the fall-through-floor bug.** It happens to everyone. Treat it as the fun puzzle, not a failure — use the "Help!" cards.
- **Celebrate the jump.** The moment a character jumps is the hook. Stop and cheer it before adding anything else.
- **Pace it:** Map + glossary $\approx$ 10 min · Steps 1–7 $\approx$ 35–45 min · that's one satisfying session. Save Part 2 for later days.

Built from the BGC Summer Camp 2026 handout (Godot 4 track, "save the reefs" platformer). The full code lives on pages 33–38 of the student handout.